

In [1]: `import pandas as pd`

`df = pd.read_csv('trends.csv')`

In [2]: `df.head()`

Out[2]:

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe

In [3]: `df.tail()`

Out[3]:

	location	year	category	rank	query
26950	Vietnam	2020	Là Gì?	1	Virus Corona là gì
26951	Vietnam	2020	Là Gì?	2	Miễn thị thực là gì
26952	Vietnam	2020	Là Gì?	3	Đầu cắt moi là gì
26953	Vietnam	2020	Là Gì?	4	Bệnh bạch hầu là gì
26954	Vietnam	2020	Là Gì?	5	Đông Lào là gì

In [4]: `df.shape`

Out[4]: (26955, 5)

In [5]: `df.columns`

Out[5]: Index(['location', 'year', 'category', 'rank', 'query'], dtype='object')

In [6]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 26955 entries, 0 to 26954
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   location    26955 non-null  object
1   year        26955 non-null  int64
2   category    26955 non-null  object
3   rank        26955 non-null  int64
4   query       26955 non-null  object
dtypes: int64(2), object(3)
memory usage: 1.0+ MB
```

In [7]: `df.dtypes`

```
Out[7]: location    object
        year       int64
        category   object
        rank       int64
        query      object
        dtype: object
```

```
In [8]: df.describe()
```

```
Out[8]:
```

	year	rank
count	26955.000000	26955.000000
mean	2015.243369	3.000000
std	3.564683	1.41424
min	2001.000000	1.00000
25%	2013.000000	2.00000
50%	2016.000000	3.00000
75%	2018.000000	4.00000
max	2020.000000	5.00000

```
In [24]: df.describe()
```

```
Out[24]:
```

	year	rank
count	26955.000000	26955.000000
mean	2015.243369	3.00000
std	3.564683	1.41424
min	2001.000000	1.00000
25%	2013.000000	2.00000
50%	2016.000000	3.00000
75%	2018.000000	4.00000
max	2020.000000	5.00000

```
In [10]: df.nunique()
```

```
Out[10]: location      83
        year          20
        category     2450
        rank           5
        query      18431
        dtype: int64
```

```
In [11]: df['category'].unique()[0:30]
```

```
Out[11]: array(['Consumer Brands', 'Men', 'Movies', 'MP3 Music Services',  
              'Music Groups', 'New Products', 'News Resources', 'Retailers',  
              'Sports', 'TV Shows', 'Video Games', 'Women', 'Athletes',  
              "Kids' Images", 'Musicians and Bands', 'News Stories',  
              'Technology', 'Travel Destinations', 'Marques', 'Personnalités',  
              "Tendances de l'année", 'Die häufigsten Suchbegriffe',  
              'Meist gesuchte Automarken', 'Meist gesuchte Musiker 2002',  
              'Searches', 'Anime Images', 'Earthquake Searches',  
              'Fictional Character Images', 'Male Celebrity Images',  
              'News Searches'], dtype=object)
```

```
In [12]: df.isnull().sum()
```

```
Out[12]: location    0  
        year        0  
        category    0  
        rank        0  
        query       0  
        dtype: int64
```

```
In [13]: (df.astype(str).apply(lambda col: col.str.strip().eq('')).sum())
```

```
Out[13]: location    0  
        year        0  
        category    0  
        rank        0  
        query       0  
        dtype: int64
```

```
In [14]: df.duplicated().sum()
```

```
Out[14]: np.int64(10)
```

```
In [23]: df[df.duplicated(keep=False)]
```

Out[23]:

	location	year	category	rank	query
19995	Kazakhstan	2018	Жылдың фильмі/ Фильм года	1	Веном
19996	Kazakhstan	2018	Жылдың фильмі/ Фильм года	2	Мстители: Война бесконечности
19997	Kazakhstan	2018	Жылдың фильмі/ Фильм года	3	Бизнес по-казахски в Америке
19998	Kazakhstan	2018	Жылдың фильмі/ Фильм года	4	Монстры на каникулах 3
19999	Kazakhstan	2018	Жылдың фильмі/ Фильм года	5	Дэдпул 2
20000	Kazakhstan	2018	Жылдың фильмі/ Фильм года	1	Веном
20001	Kazakhstan	2018	Жылдың фильмі/ Фильм года	2	Мстители: Война бесконечности
20002	Kazakhstan	2018	Жылдың фильмі/ Фильм года	3	Бизнес по-казахски в Америке
20003	Kazakhstan	2018	Жылдың фильмі/ Фильм года	4	Монстры на каникулах 3
20004	Kazakhstan	2018	Жылдың фильмі/ Фильм года	5	Дэдпул 2
25395	Kenya	2020	Trending How To (Tech)	1	How to fill Tpad online
25396	Kenya	2020	Trending How To (Tech)	2	How to create a website
25397	Kenya	2020	Trending How To (Tech)	3	How to connect a phone to TV
25398	Kenya	2020	Trending How To (Tech)	4	How to make my electric plate cooker work
25399	Kenya	2020	Trending How To (Tech)	5	How to take a screenshot on a laptop
25400	Kenya	2020	Trending How To (Tech)	1	How to fill Tpad online
25401	Kenya	2020	Trending How To (Tech)	2	How to create a website
25402	Kenya	2020	Trending How To (Tech)	3	How to connect a phone to TV
25403	Kenya	2020	Trending How To (Tech)	4	How to make my electric plate cooker work
25404	Kenya	2020	Trending How To (Tech)	5	How to take a screenshot on a laptop

```

In [27]: Q1 = df["year"].quantile(0.25)
        Q3 = df["year"].quantile(0.75)

        IQR = Q3 - Q1

        lower = Q1 - 1.5 * IQR
        upper = Q3 + 1.5 * IQR

        outliers = df[(df["year"] < lower) | (df["year"] > upper)]

        print(outliers)

```

	location	year	category	rank	query
0	Global	2001	Consumer Brands	1	Nokia
1	Global	2001	Consumer Brands	2	Sony
2	Global	2001	Consumer Brands	3	BMW
3	Global	2001	Consumer Brands	4	Palm
4	Global	2001	Consumer Brands	5	Adobe
..	...	...	...	...	...
475	Global	2005	Most Searched	1	Janet Jackson
476	Global	2005	Most Searched	2	Hurricane Katrina
477	Global	2005	Most Searched	3	Tsunami
478	Global	2005	Most Searched	4	Xbox 360
479	Global	2005	Most Searched	5	Brad Pitt

[480 rows x 5 columns]

In []: